

PROJECT ARGO

A.R.G.O



Autonomous Rover for Ground Observations

Project Brief

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0.1	9 Sep 2026	I. Mungwadzi	Initial draft: structure, scope and baseline concept
1.0	9 Sep 2026	I. Mungwadzi	First issue following naming decision (ARGO) and definition of budget, timeline and success criteria

Purpose of this Document

This brief is the founding document of the ARGO rover project. It records what is to be built, why, to what standard, within what constraints, and how the work will be organised and judged. It is written to be read in one sitting by an engineer with no prior exposure to the project, and it is the reference against which every subsequent design decision, specification and test report will be checked. Where later documents conflict with this brief, the conflict is to be resolved explicitly and this brief revised, not silently overridden.



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1. Executive Summary

ARGO is a six-wheeled, rocker-bogie autonomous rover designed, built and programmed from first principles as a single-engineer project, with a hard all-in parts budget of £300. It is a planetary-analogue vehicle: it is not intended to leave the ground it is built on, but it is designed as though it might, borrowing the architecture, discipline and verification mindset of flight rovers and applying them at hobbyist cost.

The vehicle will detect and classify obstacles using a monocular camera and an on-board neural network running on a Raspberry Pi 5, and will navigate around them without human intervention across grass, gravel, compacted soil and modest slopes. Mobility is provided by a passive rocker-bogie suspension - the same class of mechanism used on every NASA Mars rover since Sojourner - which keeps all six wheels in ground contact over uneven terrain without active control.

The project runs from September 2026 to June 2027. The headline milestone is a working autonomous prototype - driving, seeing, and avoiding - by the end of February 2027, followed by outdoor field trials and refinement through spring. The deliverables are the rover itself, a complete open design dossier (CAD, schematics, source code, bill of materials and test evidence), and a final technical report.

Success is defined quantitatively in Section 5. At minimum, ARGO must complete a 20 m outdoor course containing at least five obstacles with zero collisions in eight of ten attempts, climb a 15° slope, surmount a step of 80 mm, and operate for 30 minutes on a single charge - all for no more than £300 in parts.

At a glance

Vehicle	ARGO - six-wheel rocker-bogie rover, ~400 mm long, ≤ 5 kg
Autonomy	Camera-based object detection (YOLO-class network, on-board) with reactive obstacle avoidance
Compute	Raspberry Pi 5, Raspberry Pi OS, Python / OpenCV / NCNN inference
Terrain	Garden and parkland: grass, gravel, soil, 15–20° slopes, 80–120 mm steps
Budget	£300 hard ceiling for all purchased parts (£258 baseline + 15 % contingency)
Timeline	Sep 2026 – Jun 2027; autonomous prototype by end Feb 2027
Team	Ian Mungwadzi (sole engineer, all disciplines)
Outputs	Working rover, open design dossier, technical report, demonstration video

2. Background and Motivation

2.1 Context

Planetary rovers are among the most demanding mobile robots ever fielded: they must move over unprepared terrain, perceive hazards with limited compute, and act on that perception without a human in the loop. The engineering that makes this possible - passive suspension geometry, hazard detection, conservative autonomy - is well documented and, crucially, does not depend on a large budget to reproduce in miniature. The open-source community has demonstrated this repeatedly: NASA JPL's



Open-Source Rover and the Sawppy project have shown that a credible rocker-bogie vehicle can be built for a few hundred pounds in parts.

What those projects generally do not include is autonomy. They are, for the most part, teleoperated. Meanwhile, low-cost vision hardware has crossed a threshold: a Raspberry Pi 5 can now run a nano-class YOLO object detector at roughly 15 frames per second on its CPU alone, and dedicated on-sensor inference cameras are available for under £70. The opportunity ARGO addresses are to combine the two: a proper planetary-analogue chassis with genuine on-board perception and decision-making, at a cost within reach of any student.

2.2 Why this project

ARGO is intended to be the capstone of my mechanical engineering degree: a single artefact that demonstrates my competence across the full breadth of the discipline - structural and mechanism design, drivetrain and power, embedded electronics, software and control, systems engineering, project management and technical communication. It is deliberately scoped so that no part of it is bought as a finished solution. The frame, the suspension, the electrical architecture and the software are all to be designed, not assembled from a kit.

The project is also intended to be legible to others. Every decision is to be documented with its reasoning, every test with its evidence, and the complete design released openly so that the work can be inspected, reproduced and criticised. A rover that works is the product; a design dossier that would let someone else build one is the proof.

2.3 Naming

The vehicle is named ARGO, after the ship of the Argonauts - a vessel built for a single, difficult voyage into unknown territory, and later fixed in the sky as the constellation ARGO Navis.

3. Problem Statement

Design, build and program a mobile robot that can traverse unprepared outdoor terrain and autonomously avoid obstacles it identifies by sight, with every major subsystem - structure, mobility, power, electronics and software - designed from first principles by one engineer, for a total parts cost not exceeding £300, and with a working prototype demonstrated within six months of project start.

3.1 The core tensions

The problem is hard not because any single element is novel, but because the constraints pull against one another:

- **Cost against capability.** Six driven wheels, a suspension mechanism, a computer capable of neural-network inference, a camera, a battery and a structure must all fit inside £300 but each subsystem competes for the same money.
- **Terrain against mass.** Good off-road mobility wants large wheels, high ground clearance and torque; a tight budget and small motors want low mass and modest speeds. Suspension geometry needs to do the work that power cannot.



- **Perception against compute.** Vision-based detection on a single-board computer leaves little headroom for navigation logic, telemetry and logging. The software architecture must be disciplined about what runs where and how fast.
- **Breadth against time.** One person covering mechanical, electrical and software design in parallel with final-year study cannot afford rework.

3.2 What a solution must demonstrate

A satisfactory solution is one that can be placed on a lawn, switched on, given a heading, and trusted to reach the far side without hitting anything or getting stuck - and whose designer can then explain, with evidence, exactly why it worked.

4. Aim and Objectives

4.1 Aim

To deliver a working, fully documented, low-cost autonomous rover with all-terrain mobility and vision-based obstacle avoidance, and in doing so to produce a body of engineering work of professional standard that serves as a flagship portfolio piece.

4.2 Objectives

The aim decomposes into the following objectives. Each is traceable to one or more success criteria in Section 5.

- 01 Mobility.** Design and build a six-wheel rocker-bogie chassis, with all structural and mechanism components designed in CAD and validated by analysis, capable of the terrain performance defined in Section 5.
- 02 Powertrain and power.** Design the drive system (motor selection, gearing, wheel design, motor drivers) and the electrical power architecture (battery, regulation, protection, distribution) to achieve the required endurance and speed safely.
- 03 Electronics integration.** Design and document the complete electrical system: schematics, wiring harness, connector standard, sensor interfaces and safety cut-offs.
- 04 Perception.** Implement an on-board camera pipeline that detects and classifies obstacles in real time using a neural network, with performance characterised on a purpose-built test set.
- 05 Autonomy.** Implement navigation and obstacle-avoidance behaviour that fuses perception, odometry and inertial data to reach a goal without collision or operator input.
- 06 Verification.** Define and execute a test programme, from component bench tests to full outdoor trials, with results recorded against the success criteria.
- 07 Documentation and dissemination.** Produce a complete, openly published design dossier and a final technical report; present the project through a demonstration video and portfolio publication.
- 08 Cost and schedule discipline.** Deliver within the £300 parts ceiling and the milestone schedule in Section 9, with variances recorded and explained.



5. Success Criteria

Success is defined at three levels. “Minimum” is the threshold below which the project is not considered to have met its aim. “Target” is the design intent. “Stretch” is what would be pursued with remaining time and budget once the target is met. All criteria are measurable and will be verified by the tests listed in Section 7.5.

ID	Criterion	Minimum	Target	Stretch	Obj.
SC-01	Autonomous course completion: 20 m outdoor course, ≥ 5 obstacles, no operator input	8 / 10 runs, zero collisions	10 / 10 runs, zero collisions	10 / 10 with a moving obstacle introduced	O5
SC-02	Obstacle detection range (object ≥ 150 mm tall, daylight)	≥ 1.5 m	≥ 2.5 m	≥ 4 m, with sonar/lidar implementation	O4
SC-03	Detection throughput (end-to-end, camera to decision)	≥ 5 FPS	≥ 10 FPS	≥ 15 FPS	O4
SC-04	Detection quality on project test set (mAP@0.5, defined obstacle classes)	≥ 0.60	≥ 0.75	≥ 0.85	O4
SC-05	Slope climb, dry grass, straight ascent	15°	20°	25°	O1, O2
SC-06	Step obstacle surmounted, all six wheels	80 mm	120 mm	150 mm	O1
SC-07	Terrain classes traversed without immobilisation	Grass, gravel	Grass, gravel, soil, sand	+ loose scree, shallow mud	O1, O2
SC-08	Endurance, continuous driving on one charge	30 min	45 min	60 min	O2
SC-09	Nominal speed on flat grass	≥ 0.3 m/s	≥ 0.5 m/s	≥ 0.7 m/s	O2
SC-10	Vehicle mass, ready to run	≤ 5.0 kg	≤ 4.0 kg	≤ 3.5 kg	O1
SC-11	Dead-reckoning drift (odometry + IMU) over 20 m straight	≤ 15 %	≤ 10 %	≤ 5 %	O5
SC-12	Total purchased-parts cost	$\leq$ £300	$\leq$ £275	$\leq$ £250	O8
SC-13	Autonomous prototype demonstration date	31 Mar 2027	28 Feb 2027	31 Jan 2027	O8
SC-14	Design dossier completeness (CAD, schematics, BOM, code, tests)	All items present	All items present and cross-referenced	Independently reproducible	O7

Table 1 - Success criteria. Verification method for each is given in Table 9.

5.1 Definition of done

The project is complete when every Minimum criterion is met with recorded evidence, the design dossier is published, and the final report is issued. Meeting Target criteria is the expectation; meeting Stretch criteria is a bonus and will not be pursued at the expense of documentation quality.



6. Scope

6.1 In scope

- Complete mechanical design of chassis, rocker-bogie suspension, differential mechanism, wheel hubs and wheels, sensor and electronics mounts; fabrication by 3D printing and simple machining.
- Drivetrain design: motor and gearbox selection, torque and speed sizing, wheel design and tread, motor driver selection and thermal check.
- Electrical architecture: battery selection, regulation, fusing, master cut-off, wiring harness, connector standard, sensor interfaces, schematics and as-built wiring diagrams.
- On-board software: operating-system image, camera pipeline, neural-network inference, obstacle-avoidance behaviour, motor control, telemetry, logging and a remote emergency stop.
- Perception dataset: collection and labelling of a project-specific image set for evaluation (and, if required, fine-tuning) of the detector.
- Ground-station software: a laptop-side telemetry dashboard and manual override interface over Wi-Fi.
- Verification: bench tests, mobility tests, perception evaluation and full autonomous field trials, with a test report.
- Documentation: this brief, a requirements specification, design review packs, the design dossier, a build log and the final report.

6.2 Out of scope

- Operation in any extraterrestrial or vacuum environment; radiation, thermal-vacuum or launch-load qualification. ARGO is a terrestrial analogue only.
- Global localisation and mapping (GNSS, SLAM, occupancy grids). Navigation is local and reactive; a waypoint capability via dead reckoning is a stretch goal, not a requirement.
- Manipulation: no robotic arm, sampling tools or science instruments in this phase.
- Weatherproofing beyond light rain resistance and dust tolerance; no submersion or extreme-temperature operation.
- Custom printed-circuit-board design. Off-the-shelf breakout boards are used; a bespoke PCB may be considered in a future phase.
- Night operation and active illumination.
- Multi-rover coordination or any form of swarm behaviour.

6.3 Constraints

Constraint	Value	Rationale
Budget	£300 hard ceiling, all purchased parts	Defines the project; forces design over procurement



Constraint	Value	Rationale
Schedule	Autonomous prototype by 28 Feb 2027; close-out by June 2027	Aligns with final year and leaves time for trials and reporting
Team	One engineer	All disciplines covered by me; drives simplicity
Compute	Raspberry Pi 5 class, no discrete GPU	Cost; all inference must be CPU- or sensor-side
Fabrication	FDM 3D printing (PLA/PETG), hand tools, access to university workshop	No CNC or composite tooling assumed
Power	Single 3S lithium-polymer pack, ≤ 12.6 V	Cost and availability; imposes safety requirements
Size	Fits in a $450 \times 350 \times 300$ mm envelope	Transportable by hand and on public transport
Environment	UK garden/parkland, daylight, dry to lightly damp conditions	Test venue availability

Table 2 - Project constraints.

6.4 Assumptions

- I have access to, or can borrow, a laptop, soldering equipment, hand tools, a lithium-polymer charger and access to a 3D printer from the university; these are not charged to the project budget.
- Component prices are indicative as of September 2026 and will be refreshed at the procurement milestone. Single-board-computer pricing is currently volatile.
- Open-source software (Raspberry Pi OS, OpenCV, Ultralytics YOLO, NCNN) will remain available under their current licences.
- A suitable outdoor test area (grass, gravel, a modest slope) is accessible within a short distance throughout the trials period.

7. Proposed Methodology

7.1 Approach

ARGO will be developed using a lightweight systems-engineering process modelled on the V-model: requirements are defined before concepts, concepts are traded before detailed design, and every design output has a corresponding verification activity planned at the time it is designed. Three formal review gates - Concept Design Review, Critical Design Review and Test Readiness Review - mark the transitions between phases. Because the project has one engineer (myself), these reviews are self-conducted against written checklists, with the review pack published in the design dossier so that the reasoning is open to external scrutiny.

Within each phase, work is organised by subsystem so that mechanical, electrical and software streams can advance in parallel and integrate at defined points. Long-lead items (motors, single-board computer, camera) are ordered at the Critical Design Review so that the integration phase is not blocked by procurement.



7.2 Design principles

The following principles govern every design decision. Where two options are otherwise equal, the one that better satisfies the earlier principle wins.

1. **Design, Don't Buy.** Structure, mechanisms, electrical architecture and software are original work. Commercial parts are used at the component level (motors, boards, sensors) - not as pre-integrated subsystems or kits.
2. **Passive before active.** Prefer geometry and mechanism over control and computation. For example, the rocker-bogie keeps wheels on the ground without a single line of code.
3. **Conservative autonomy.** When uncertain, stop. The vehicle must never trade a collision for progress. A safe halt is always an acceptable outcome; a collision never is.
4. **Every requirement has a test.** No requirement is written without a stated method of verifying it, and no test is run without recording the result against the requirement.
5. **Reproducible by a stranger.** The dossier is written so that a competent engineer with no contact with me could build a second ARGO from it.
6. **Spend on whatever moves the metric.** Budget goes to the components that most directly drive the success criteria: motors, camera, compute. Everything else is printed, made or scrounged.

7.3 Phases and review gates

Phase	Activities	Exit gate / evidence
0 - Initiation	Issue this brief; create repository and dossier structure; set up build log; confirm test venue and workshop access.	Brief v1.0 issued; repository live
1 - Requirements & concept	Requirements specification derived from Section 5; trade studies on suspension configuration, steering method, drive motor class, perception hardware and software stack; system architecture; mass and power budgets (first cut).	Concept Design Review (CoDR): concept frozen, budgets allocated per subsystem
2 - Detailed design	Full CAD of chassis and mechanisms; hand calculations and FEA on rocker, bogie and differential members; wheel and hub design; electrical schematics and harness drawing; software architecture and interface definitions; final BOM.	Critical Design Review (CDR): drawings released, BOM ordered, test plan drafted
3 - Procurement & fabrication	Order all parts; print and finish structural components; machine axles and shafts; assemble suspension and drivetrain; bench-test motors and drivers.	Chassis rolling under manual push; drivetrain bench-tested
4 - Integration	Install power, compute and sensors; harness assembly; motor control software; teleoperation over Wi-Fi; telemetry dashboard; first driven test indoors.	First Drive: rover driven under remote control on flat ground
5 - Autonomy	Camera pipeline and detector deployment; dataset capture and evaluation; obstacle-avoidance behaviour; indoor obstacle course; safety cut-offs proven.	Test Readiness Review (TRR) and Autonomous Demonstration (the working prototype)
6 - Field trials & refinement	Outdoor terrain campaign against SC-01 to SC-11; failure analysis; design iterations on wheels, tread and behaviour tuning; endurance runs.	Test report: all Minimum criteria met with evidence



Phase	Activities	Exit gate / evidence
7 - Close-out	Final report; dossier completion and publication; demonstration video; lessons learned; project retrospective.	Final report issued; dossier public

Table 3 - Development phases. Dates are given in the schedule, Section 9.

7.4 Subsystem approach

7.4.1 Structure and mobility

The baseline is a six-wheel rocker-bogie with a body-mounted differential (a bar-and-link differential in the first instance, for cost and ease of printing), giving passive terrain-following across all six wheels. Steering is by differential drive - skid steer - in the baseline configuration, which eliminates four steering actuators and their cost, mass and failure modes; corner-wheel steering is a stretch option to be reconsidered after field trials. Structural members will be FDM-printed in PETG with aluminium tube or rod where stiffness demands it. Wheel diameter is targeted at 100–120 mm, which sets the achievable step height; tread and compliance will be iterated during trials.

7.4.2 Powertrain and power

Six brushed DC gear motors with integrated encoders, one per wheel, driven by dual H-bridge driver modules. Motor sizing will be by a traction and gradeability calculation on the worst-case terrain (20° slope, 4 kg vehicle, 50 % traction coefficient) with a 1.5× margin. Power comes from a single 3S lithium-polymer pack through a fused distribution point: a 5 V/5 A regulator for compute and sensors, and direct battery voltage to the motor drivers. A hardware master switch and a software-controlled motor-enable line provide two independent means of stopping the drive.

7.4.3 Electronics and sensing

Compute is a Raspberry Pi 5. Perception uses a Raspberry Pi Camera Module 3 (wide) on the CSI interface, mounted on a fixed mast. Proprioception uses wheel encoders and a 9-axis inertial measurement unit over I²C. A forward-facing time-of-flight rangefinder provides an independent, low-latency near-field stop that does not depend on the neural network - a deliberate redundancy in line with the conservative-autonomy principle. All connectors follow a single documented standard (JST-XH for signal, XT30 for power) so that the harness is maintainable.

7.4.4 Software and autonomy

The software runs on Raspberry Pi OS Lite and is written in Python, with performance-critical inference handled by a nano-class YOLO detector exported to the NCNN runtime, which is the fastest CPU path currently available on the Pi 5. Published benchmarks give roughly 15 FPS for a nano model at 640-pixel input on the Pi 5 CPU; running at 320–416 pixels are expected to comfortably exceed the 10 FPS target while leaving headroom for control. The architecture is a set of cooperating processes - camera and detection, state estimation, behaviour, motor control, telemetry - communicating over lightweight local messaging. Obstacle avoidance is reactive: detections are projected into a body-frame hazard map and steering is chosen by a vector-field-histogram-style method, with the time-of-flight sensor overriding to a halt. The Robot Operating System (ROS 2) was considered and rejected for the baseline as disproportionate to the compute available and the size of the codebase; the interface design will not preclude a later migration.



7.4.5 Ground station

A laptop-side dashboard over Wi-Fi displays live telemetry (battery, motor currents, detections, behaviour state), provides manual override and a software emergency stop, and records every run for post-test analysis. Telemetry logs are the primary evidence base for the test report.

7.5 Verification strategy

Verification is planned at four levels. Each success criterion is mapped to a test in Table 9 (Section 12). The principle is that nothing is “tested” by demonstration alone; every test has a written procedure, pass criteria, and a recorded result.

- **Analysis.** Hand calculations and FEA for structural members; traction and gradeability calculations for motor sizing; power budget for endurance.
- **Bench test.** Motor torque-speed characterisation; regulator load test; detector throughput and accuracy on the project dataset; sensor latency.
- **Subsystem test.** Chassis on a test rig for suspension articulation; drivetrain under load; perception pipeline in the loop with recorded video.
- **Field trial.** Full vehicle on the defined outdoor course, repeated runs, all telemetry logged, against SC-01 to SC-11.

8. System Concept and Technical Baseline

This section records the baseline that the trade studies in Phase 1 will confirm or overturn. It is a starting point, not a decision; its purpose is to make the budget and schedule estimates concrete.

8.1 Architecture summary

Configuration	Six-wheel rocker-bogie, body-mounted differential, skid steer
Dimensions (target)	$\approx 400 \times 300 \times 250$ mm; wheel $\varnothing$ 100–120 mm; ground clearance ≥ 60 mm
Mass (target)	≤ 4.0 kg ready to run
Drive	6×12 V brushed DC gear motors with encoders; $3 \times$ dual H-bridge drivers
Power	3S 2200 mAh LiPo (11.1 V nominal); fused; 5 V/5 A buck for compute
Compute	Raspberry Pi 5 with active cooler; Raspberry Pi OS Lite
Perception	Camera Module 3 (wide) $\rightarrow$ YOLO nano (NCNN) at 320–416 px
Proprioception	$6 \times$ wheel encoders; 9-axis IMU (I ² C)
Safety sensing	Forward ToF rangefinder; hardware master switch; software motor-enable
Comms	Wi-Fi to laptop ground station; telemetry + override + E-stop
Software	Python; OpenCV; Ultralytics YOLO export; multiprocessing architecture



8.2 Key trade studies (Phase 1)

Trade	Options	Baseline	Decision drivers
Steering	Skid steer 4-corner steering	Skid steer	Cost (4 servos), mass, complexity; turf damage; turning radius
Differential	Bar-link Gear differential	Bar-link	Printability, backlash, cost
Perception hardware	Camera Module 3 + CPU inference AI Camera (IMX500, on sensor) Stereo/depth	Camera 3 + CPU	Custom-model flexibility vs. CPU headroom; £25 vs £67; depth cost
Compute memory	Pi 5 2 GB 4 GB 8 GB	2 GB	Price volatility; nano-model footprint is small; headless OS
Motor class	N20 micro gear 25 mm 12 V gear 37 mm 12 V gear	25 mm 12 V	Torque for 20° slope at 4 kg vs. mass and cost
Wheel construction	Printed rigid + rubber tread Printed TPU compliant Commercial	Printed + TPU tread	Traction, cost, repairability
Middleware	Plain Python multiprocessing ROS 2	Multiprocess	Pi 5 headroom, learning curve, codebase size

Table 4 - Phase 1 trade studies and current baseline.

9. Work Breakdown and Schedule

The schedule runs across the academic year. It is front-loaded so that the design is frozen before the December break, procurement happens before the January price and stock risk of the new year, and the autonomous prototype is demonstrated with a full term of trials still available. Effort assumes roughly eight to twelve hours per week alongside study, both rising during vacation periods.

Phase	Period	Milestone	Date
0 - Initiation	Sep 2026	M0 Brief issued; repository live	15 Sep 2026
1 - Requirements & concept	Sep – Oct 2026	M1 Concept Design Review	31 Oct 2026
2 - Detailed design	Nov 2026	M2 Critical Design Review; parts ordered	30 Nov 2026
3 - Procurement & fabrication	Dec 2026	M3 Rolling chassis; drivetrain bench-tested	31 Dec 2026
4 - Integration	Jan 2027	M4 First Drive (teleoperated)	31 Jan 2027
5 - Autonomy	Feb 2027	M5 Autonomous Demonstration - working prototype	28 Feb 2027
6 - Field trials & refinement	Mar – Apr 2027	M6 Test report: all Minimum criteria met	30 Apr 2027
7 - Close-out	May – Jun 2027	M7 Final report and dossier published	15 Jun 2027

Table 5 - Milestone schedule. M5 is the headline milestone.



9.1 Schedule logic

- Design work (Phases 1–2) is scheduled during term, when workshop access is available for measurement and prototyping, but fabrication time is limited.
- Fabrication (Phase 3) is placed over the winter vacation, when sustained bench time is available.
- Integration and autonomy (Phases 4–5) are each given a full month with a single, unambiguous exit milestone, so that slippage is visible immediately.
- Two months of trials (Phase 6) provide slack: if M5 slips by up to four weeks, the Minimum success-criteria evidence can still be gathered before close-out. A slip beyond 31 March 2027 triggers descoping (see risk R-01).
- Close-out (Phase 7) is protected. Documentation is not to be compressed to recover schedule elsewhere.

10. Budget

The budget covers every purchased part on the vehicle. Tools, consumables already owned, the laptop, the battery charger and 3D-printer access are excluded (Section 6.4). Prices are indicative September 2026 UK retail including VAT, rounded to the pound; they will be refreshed at the CDR when the order is placed. A 15 % contingency is held against price movement and breakage, and is released only by a recorded decision.

Subsystem	Item	Qty	Indicative cost (£)
Compute	Raspberry Pi 5 (2 GB)	1	52
	Active cooler	1	5
	microSD card, 64 GB, A2	1	8
Perception	Raspberry Pi Camera Module 3 (wide)	1	25
	Time-of-flight rangefinder (VL53L1X class)	1	8
Navigation sensing	9-axis IMU breakout (ICM-20948 class)	1	8
Mobility	12 V 25 mm gear motors with encoders	6	42
	Wheels: printed hubs, TPU tread, bearings	6	15
	Shafts, bearings, fasteners, inserts	-	12
Drive electronics	Dual H-bridge driver modules (TB6612FNG class)	3	15
	5 V / 5 A buck regulator	1	8
	Wiring, connectors, fuse holder, master switch	-	10
Power	3S 2200 mAh LiPo with XT60	1	20
Structure	PETG / TPU filament	≈ 1.5 kg	18
	Aluminium tube and rod	-	12



Subsystem	Item	Qty	Indicative cost (£)
Baseline subtotal			258
Contingency	15 % of baseline		39
Total (ceiling £300)			297

Table 6 - Indicative bill of materials and budget.

10.1 Budget notes

- Compute and perception together are 38 % of the baseline. If Raspberry Pi pricing moves further, the Phase 1 trade study will reassess the AI Camera (on-sensor inference, $\approx$ £67) paired with a lower-cost board, which is cost-neutral overall but changes the software approach.
- Motors are the single largest line and the item most likely to need a second purchase after trials; a spare pair is the first call on contingency.
- Corner steering (4 servos, $\approx$ £25–40) is not budgeted. If Target criteria are met with margin, it may be funded from contingency as a stretch item.
- All spend is logged against this table in the dossier, with supplier, date and actual price.

11. Risk Register

Risks are scored on likelihood (L) and impact (I) from 1 (low) to 5 (high). The register is reviewed at each gate and whenever a risk materialises. Only risks scoring 8 or more are listed here; the full register lives in the dossier.

ID	Risk	L	I	Score	Mitigation / response
R-01	Schedule slip: autonomy milestone M5 missed because of study workload	4	4	16	Front-loaded design; monthly milestones; descoping ladder pre-agreed: drop SC-11 first, then reduce terrain classes, then accept M5 on 31 Mar
R-02	Insufficient traction or torque: rover fails 15° slope or 80 mm step	3	4	12	Size motors with 1.5× margin; TPU tread iteration budgeted; wheel diameter can grow to 120 mm within envelope
R-03	Detector performance inadequate outdoors (lighting, grass clutter)	3	4	12	Collect project dataset early (Phase 4); fine-tune nano model; ToF sensor guarantees safe stop regardless
R-04	Compute headroom: inference starves control loop	3	3	9	Run detector at 320–416 px; separate processes with priorities; hard real-time functions (E-stop, ToF) independent of vision
R-05	Component price rise or stock-out at procurement	3	3	9	Order at CDR; 15 % contingency; pre-identified substitutes for compute and camera
R-06	Printed structural part failure under load	3	3	9	FEA on rocker/bogie members; print orientation specified on drawings; aluminium reinforcement where flagged



ID	Risk	L	I	Score	Mitigation / response
R-07	Lithium-polymer battery incident (over-discharge, short, physical damage)	2	5	10	Fuse at pack; low-voltage cut-off in software and alarm; charging only in a LiPo bag under supervision; pack enclosed and strain-relieved
R-08	Single-person dependency: illness or other commitment stalls all streams	2	4	8	Weekly build log so state is always recoverable; protected close-out phase; documentation continuous, not deferred
R-09	Scope creep toward mapping, manipulation or ROS migration	3	3	9	Out-of-scope list (Section 6.2) is binding; changes require a revision to this brief

Table 7 - Risk register (score ≥ 8).

12. Deliverables and Verification Matrix

12.1 Deliverables

ID	Deliverable	Description	Due
D-01	Project brief	This document	M0
D-02	Requirements specification	Numbered, verifiable requirements traced to Section 5	M1
D-03	Concept design review pack	Trade studies, architecture, mass and power budgets	M1
D-04	Critical design review pack	Released CAD and drawings, analysis reports, schematics, harness drawing, final BOM, test plan	M2
D-05	ARGO rover	The vehicle, as built, matching the released design	M5
D-06	Source code repository	On-board software, ground station, dataset tools, with README and licence	M5 (ongoing)
D-07	Perception dataset and evaluation	Labelled test set, evaluation script, results	M5
D-08	Test report	Procedures, results and evidence for every success criterion	M6
D-09	Design dossier	Complete, cross-referenced package of D-02 to D-08 plus build log, published openly	M7
D-10	Final technical report	Formal report: design, analysis, results, discussion, lessons learned	M7
D-11	Demonstration video	Short film of ARGO completing the autonomous course, with technical narration	M7

Table 8 - Deliverables.

12.2 Verification matrix

Criterion	Verification method	Level	Evidence
SC-01	Ten timed runs on the defined course; collisions counted by observer and telemetry	Field trial	Run log, video, telemetry
SC-02	Static target at measured distances; first-detection distance recorded	Subsystem	Detection log



Criterion	Verification method	Level	Evidence
SC-03	Pipeline timing instrumentation over 5-minute run	Bench	Timing histogram
SC-04	Evaluation script on held-out project test set	Bench	mAP report
SC-05	Inclined ramp at measured angle; straight ascent from rest	Field trial	Video, inclinometer
SC-06	Stacked step of measured height; all wheels over	Subsystem	Video
SC-07	Traverse of each terrain patch; immobilisation counted	Field trial	Run log
SC-08	Continuous figure-of-eight until low-voltage cut-off	Field trial	Battery telemetry
SC-09	Timed 10 m straight on flat grass	Field trial	Stopwatch, encoder log
SC-10	Weighed ready to run	Bench	Scale photo
SC-11	Commanded 20 m straight; end position measured against dead-reckoned	Field trial	Survey tape, pose log
SC-12	Purchase ledger reconciled	Analysis	Ledger
SC-13	Date of M5 demonstration	Analysis	Build log entry, video
SC-14	Dossier checklist audit	Analysis	Checklist

Table 9 - Verification matrix mapping success criteria to tests.

13. Stakeholders, Audience and Governance

13.1 Stakeholders and audience

Stakeholder	Interest	What they need from the project
Ian Mungwadzi (author)	Engineer, owner, primary user	A working vehicle and a body of work that demonstrates professional competence across the discipline
Prospective employers and assessors	Evaluators of the portfolio	Evidence of rigour: clear reasoning, measurable results, honest reporting of what did not work
Open-source and maker community	Potential builders and critics	A reproducible dossier, permissive licensing, clear documentation
Test-venue owners and bystanders	Safety	A vehicle that stops reliably and does no damage to bystanders and/or surroundings

Table 10 - Stakeholders.

13.2 Governance

The project is governed by this brief and the gate reviews. I act as both engineer and reviewer; to keep that honest, every gate review is conducted against a published checklist, its outcome is written up in the dossier, and open actions are tracked to closure. Changes to scope, budget ceiling or milestone dates require a revision to this brief with the change recorded in the revision history. Progress is recorded in a weekly build log, which doubles as the recovery point if work is interrupted.



13.3 Reporting cadence

- Weekly: build log entry (what was done, what was learned, what is next).
- Monthly: milestone check against Table 5; risk register review.
- At each gate: review pack published; go / no-go recorded.

14. Quality, Safety and Ethics

14.1 Quality

Quality is defined here as the traceability of every artefact: each requirement to a test, each drawing to a part, each part to a purchase, each test to a result. Drawings carry revision letters; software is version-controlled with tagged releases at each gate; the BOM records as-designed and as-built states. The intent is that the final report can be audited from the dossier without reference to my memory.

14.2 Safety

- Lithium-polymer handling: charging in a fire-resistant bag, never unattended; pack inspected before every session; low-voltage cut-off enforced in software with an audible alarm; pack mechanically enclosed and strain-relieved on the vehicle.
- Electrical: fuse at the pack; master switch accessible from outside the vehicle; regulator and driver thermal check under sustained load before field use.
- Mechanical: no exposed pinch points at the differential; wheel fasteners torque-checked before every trial.
- Operational: software emergency stops from the ground station and an independent hardware stop; trials in areas clear of people and animals; the vehicle is never left running unattended.
- Workshop: university workshop rules and inductions followed for all machining; eye protection when cutting or drilling; solder-fume extraction.

14.3 Ethics

The design will be published under permissive licences (hardware under CERN-OHL-P or equivalent; software under MIT) so that the work can be inspected, reproduced and improved. Third-party components, models and datasets will be credited and their licences honoured. Test results will be reported as measured, including failures; no criteria will be reported as met without the evidence listed in Table 9. Images collected for the perception dataset will exclude identifiable people.

Appendix A - Glossary

Term	Meaning
Rocker-bogie	Six-wheel passive suspension in which each side has a rocker arm carrying a two-wheel bogie; the two rockers are linked through a differential, so the body tilts by the average of the two sides



Term	Meaning
Differential (mechanism)	The linkage that averages rocker motion between sides; in ARGO, a bar-and-link mechanism across the top of the body
Skid steer	Steering by driving the two sides at different speeds, with no steerable wheels
YOLO	A family of single-pass neural-network object detectors; the “nano” variants are sized for embedded devices
NCNN	A neural-network inference runtime optimised for ARM CPUs
mAP@0.5	Mean average precision at 50 % intersection-over-union; a standard object-detection accuracy metric
IMU	Inertial measurement unit: accelerometer, gyroscope and, here, magnetometer
ToF	Time-of-flight rangefinder; a short-range laser distance sensor
CoDR / CDR / TRR	Concept Design Review, Critical Design Review, Test Readiness Review - the project’s gate reviews
Dossier	The complete published design package (Section 12)

Appendix B - References

1. NASA Jet Propulsion Laboratory, Open-Source Rover Project (github.com/nasa-jpl/open-source-rover). Reference design for an educational rocker-bogie rover.
2. R. Cheng, Sawppy the Rover: a motorised model of Curiosity and Perseverance for under \$500 in parts (github.com/Roger-random/Sawppy_Rover).
3. Ultralytics, Raspberry Pi deployment guide and benchmarks (docs.ultralytics.com/guides/raspberry-pi). Source for nano-model inference timings on Raspberry Pi 5.
4. Raspberry Pi Ltd, Raspberry Pi 5 and Camera Module 3 product documentation (raspberrypi.com/documentation).
5. Raspberry Pi Ltd, AI Camera (IMX500) documentation (raspberrypi.com/documentation/accessories/ai-camera.html).
6. J. Borenstein and Y. Koren, “The Vector Field Histogram - fast obstacle avoidance for mobile robots,” IEEE Transactions on Robotics and Automation, vol. 7, no. 3, 1991.
7. D. Harrington and C. Voorhees, “The challenges of designing the rocker-bogie suspension for the Mars Exploration Rover,” 37th Aerospace Mechanisms Symposium, NASA, 2004.

